

Fixed-M SMLM Localization Report — CNN+KAN v2

1. Network Architecture and Changes (v1 → v2)

The localizer is a hybrid CNN + KAN model (not KAN alone). One separate model is trained for each emitter count $M = 1, 2, \dots, 10$. At inference, a CNN M-classifier routes each 16×16 px frame to the correct fixed-M localizer, followed by PSF template-matching refinement.

Inference pipeline

- 16×16 frame → CNN M-classifier ($M \in \{1, \dots, 10\}$)
- → CNN+KAN localizer `kan_M{M}.pt`
- → PSF refinement → predicted (x, y) in nm

v2 architecture (FixedMKANLocalizerV2)

- CNN spatial encoder: preserves full 16×16 feature map (no global average pooling).
- Candidate scoring: CNN features gathered at each of 80 allowed pixel centers; per-slot MLP scores all candidates.
- Slot embeddings: separate head per emitter slot (M heads).
- FastKAN offset head: sub-pixel ($\Delta x, \Delta y$) from global + local context.
- Decode: softmax over 80 centers + tanh-bounded offset → position in nm.

Key training improvements in v2

- Hungarian matching loss for $M \geq 2$.
- Higher coordinate loss weight; CE warmup then joint training.
- Spatially aware head critical for $M \geq 4$ overlapping PSFs.
- Trained on GPU (NVIDIA RTX 4060 Laptop, CUDA, mixed precision).

v1 vs v2 summary

Component	v1	v2
Spatial encoder	Global avg pool → 1024-d vector	Full 16×16 feature map
Localization	Global vector → 80-way softmax	Candidate-aligned feature gathering
Slot loss	Fixed emitter order	Hungarian matching ($M \geq 2$)
Offset head	Linear only	FastKAN + linear heads

2. Test Datasets — Simulated vs Real

A) Simulated test (combined_test/) — quantitative RMSMD

- 500 held-out simulated frames (50 per M, $M=1 \dots 10$), generated independently from training data.
- Ground truth available → RMSMD computed (MATLAB RMSMD(S,X)).
- Full 16×16 px frames, no window cropping.
- Pipeline: CNN M-classifier → CNN+KAN → PSF refinement.
- M detection accuracy: 92.8%.

B) Real test — EPFL public dataset (blind, qualitative)

- Dataset: EPFL SMLM 2D Challenge — Bundled Tubes Long Sequence
- 30 frames processed; 272 localizations predicted.
- Ground truth NOT used → RMSMD cannot be computed (blind test).
- Method: peak detection → 16×16 patches → CNN M-classifier → CNN+KAN → refinement.
- Output: super-resolution reconstruction image (see Section 4).

3. RMSMD Results — Simulated Test (all M)

Metric: mean RMSMD in nm over 50 test frames per M. Oracle M: true emitter count given.

Pipeline: CNN predicts M.

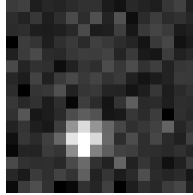
M	v1 oracle (nm)	v2 oracle (nm)	v2 pipeline (nm)
1	9.2	8.8	8.8
2	37.7	41.6	41.6
3	74.7	49.5	49.5
4	98.9	66.1	64.7
5	105.6	66.6	66.6
6	127.5	68.3	68.3
7	172.6	63.4	64.8
8	158.1	63.8	63.5
9	179.4	84.4	84.7
10	189.3	119.8	112.5
Overall (500)	115.3	63.2	62.5

Overall improvement: mean RMSMD reduced from 115.3 nm (v1) to 63.2 nm (v2) — approximately 45% reduction.

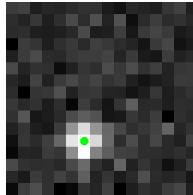
4. Example Simulated Training Frames

Sample 16×16 px simulated frames (grayscale, upscaled, no axis labels). Green = ground truth, red = prediction (overlay folder: example_frames_overlay/).

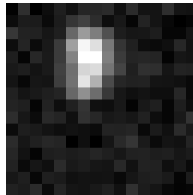
- $M = 1$



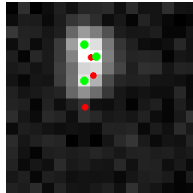
With GT (green) and prediction (red):



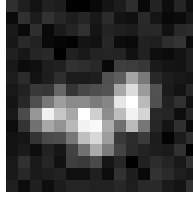
- $M = 3$



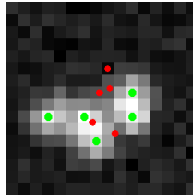
With GT (green) and prediction (red):



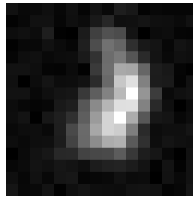
- $M = 5$



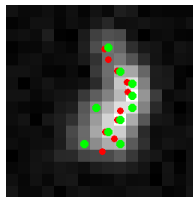
With GT (green) and prediction (red):



- $M = 10$



With GT (green) and prediction (red):



I use a CNN+KAN hybrid with per-M models and a CNN M-classifier. On simulated blind test, mean RMSMD improved from ~ 115 nm to ~ 63 nm. On the public EPFL real SMLM dataset, I report localizations and a super-resolution reconstruction (no RMSMD — ground truth not used).

KAN final reconstruction — all 200 EPFL frames

